Generative AI–Driven Scenario Generation and Stochastic Optimization for Solar-Powered Green Hydrogen Systems

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Chapter title: Generative AI–Driven Scenario Generation and Stochastic Optimization for Solar-Powered Green Hydrogen Systems

Target part: Part III – Generative AI for Renewable Energy Forecasting, Scenario Generation, and Uncertainty Modeling (also fits Part II – Stochastic, Robust, and Uncertainty-Aware Optimization)

Abstract (5 sentences): Sizing solar-powered green hydrogen plants on a single typical weather year hides the bad years that drive storage needs and supply shortfalls. This chapter couples a physics-informed conditional variational autoencoder, trained on open NREL irradiance data, with a genetic algorithm that sizes the PV array, PEM electrolyser, battery and hydrogen tank under risk-neutral and CVaR risk-averse formulations with a reliability chance constraint. The generator produces realistic hourly weather years with multi-day persistence and matches or outperforms Gaussian and bootstrap baselines on probabilistic scores. In semi-arid and humid case studies, scenario-based designs kept 95 % supply reliability in 49–78 % of unseen real-weather years versus 32–42 % for single-year designs, for an LCOH premium of only 1–2.5 %. The chapter closes with practical lessons on validating generative models for storage sizing, arid-site deployment and trustworthy AI in hydrogen infrastructure.

Abstract

Sizing a solar-powered green hydrogen plant on a single typical weather year hides the bad years that drive storage needs and supply shortfalls. This chapter couples generative AI with mathematical optimisation to design for the full range of weather instead. A physics-informed conditional variational autoencoder (CVAE), trained on open NREL irradiance data, generates hourly clearness-index and temperature profiles on a clear-sky envelope, and a latent autoregressive process strings them into synthetic years with

realistic multi-day persistence. A real-coded genetic algorithm then sizes PV, PEM electrolyser, battery and hydrogen tank under deterministic, risk-neutral and CVaR risk-averse formulations with a chance constraint on 95 % supply reliability. In a semi-arid and a humid case study, the CVAE matched or outperformed Gaussian and bootstrap baselines, and scenario-based designs kept the reliability target in 49–78 % of unseen real-weather years versus 32–42 % for single-year designs, at an LCOH premium of only 1–2.5 % (7.8–8.1 USD/kg). The chapter closes with lessons on validating generative models for storage sizing, arid-site deployment and trustworthy AI in hydrogen infrastructure.

Keywords: green hydrogen · generative AI · variational autoencoder · scenario generation · genetic algorithm · stochastic optimisation · CVaR · PEM electrolyser · solar energy

1 Introduction

Green hydrogen made by water electrolysis from solar power is a cornerstone of deep decarbonisation for steel, ammonia, refining and heavy transport [1, 2]. Its cost, however, is dominated by two coupled design decisions: how much renewable generation and electrolyser capacity to install, and how much electrical and hydrogen storage to add to ride through variability. Both decisions depend on the weather the plant will see over a 20–25-year life — not on the single "typical" year usually used for design.

Classical design studies size the plant against one historical or typical meteorological year (TMY). A TMY is built to represent average conditions; by construction it under-represents long cloudy spells and bad years, exactly the events that drive storage size and supply shortfalls. Stochastic programming addresses this by optimising over many weather scenarios [3], but good scenarios are hard to get: on-site measurements rarely cover more than a few years, and simple statistical models (Gaussian, autoregressive) struggle with the bounded, diurnal, non-Gaussian nature of solar irradiance.

Generative artificial intelligence offers a way out. Deep generative models such as variational autoencoders (VAEs) [4] and generative adversarial networks (GANs) [5] learn the joint distribution of whole daily profiles and can draw an unlimited number of new, realistic scenarios from a short record [6]. What is still missing is a transparent, reproducible link between such a generator and the sizing optimisation of an off-grid hydrogen plant, and an honest comparison against simple baselines.

This chapter closes that gap with an end-to-end, open-data workflow. Its contributions are:

1. A **physics-informed conditional VAE** (CVAE) that generates hourly irradiance and temperature profiles as a *clearness index* times a clear-sky envelope, so every synthetic day respects sunrise, sunset and the seasonal irradiance ceiling by construction.

2. A **latent first-order autoregressive (AR(1)) process** that strings generated days into full synthetic years with realistic multi-day persistence of cloudy and sunny weather.
3. A **genetic-algorithm (GA) based stochastic sizing** of a PV–battery–PEM electrolyser–hydrogen-tank system with a rule-based energy management system, a chance constraint on supply reliability and a risk-averse CVaR variant.
4. A **two-climate case study** (a semi-arid high-desert site and a humid subtropical site) using open NREL data, with out-of-sample evaluation on both generated and bootstrapped weather years.

All code is written from scratch in Python/NumPy and released with the chapter. Section 2 reviews related work; Section 3 models the hydrogen system; Section 4 presents the data and the generative model; Section 5 formulates the optimisation; Section 6 reports results; Section 7 discusses operational trends and challenges; Section 8 concludes.

2 Background and Related Work

Green hydrogen systems. A solar-fed hydrogen plant couples photovoltaic (PV) generation, an electrolyser (alkaline or proton-exchange membrane, PEM), compression and storage [7, 8]. PEM electrolysers suit variable power because they ramp in seconds and operate down to roughly 5–10 % of rated load, at the price of higher capital cost [8, 9]. The levelised cost of hydrogen (LCOH) is driven mostly by electrolyser capital cost, its capacity factor and the cost of electricity [1]; storage decides how much of the solar resource can be turned into a steady hydrogen supply.

Optimisation of hydrogen system design. Capacity planning is usually posed as a mixed-integer linear programme over one representative year, or solved with metaheuristics when the operating model is non-linear. Genetic algorithms [10] and their multi-objective form NSGA-II [11] are popular because they treat the plant simulator as a black box, so part-load efficiency curves, minimum-load limits and rule-based control can be kept exactly. Stochastic programming [3] and risk measures such as conditional value-at-risk (CVaR) [12] extend these designs to uncertain weather, but they need many representative scenarios, and scenario-reduction methods [13] can only reduce what already exists.

Scenario generation with generative AI. Chen et al. [6] showed that GANs can generate wind and solar power scenarios that match the statistics of historical data without an explicit probability model. VAEs [4] offer a stable maximum-likelihood alternative with an explicit latent space, and diffusion models [14] have since raised sample quality further. Physics-informed learning [15] suggests embedding known structure — here, the clear-sky envelope — instead of asking the network to learn it. Generated scenarios are judged with proper scoring rules such as the energy score and the continuous ranked probability score (CRPS) [16].

Gap. Most generative-scenario studies stop at forecast-style scores, and most hydrogen sizing studies use a single year. Few connect the two, check whether the generator beats simple baselines, or test the resulting design on weather it was not optimised on. This chapter does all three on open data.

3 Green Hydrogen System Modelling

The plant is off-grid: PV feeds a PEM electrolyser directly, a lithium-ion battery shifts surplus PV into the night, and a compressed-gas tank buffers hydrogen against a constant industrial demand of 100 kg/day. Any hydrogen the plant cannot supply is bought in by truck at a penalty price. The model runs hourly over 8 760 h.

3.1 Photovoltaic generator

PV output per installed kWp follows a temperature-corrected linear model with the NOCT cell-temperature approximation:

$$P_{pv}(t) = C_{pv}\,\eta_{d}\,\frac{G(t)}{1000}\,\bigl[1 + \gamma\,(T_c(t) - 25)\bigr], \qquad T_c(t) = T_a(t) + G(t)\,\frac{NOCT - 20}{800}$$

with plane-of-array irradiance G (W/m²), air temperature T_a (°C), derate η_d = 0.86, γ = −0.4 %/°C and NOCT = 45 °C. The array faces south at a tilt equal to the site latitude; G is obtained from (generated) horizontal irradiance by the Erbs diffuse-fraction model [22] and an isotropic-sky transposition with ground albedo 0.2, using pvlib sun positions [17].

3.2 PEM electrolyser and compression

The electrolyser runs between a minimum load of 10 % and its rating C_el. Specific energy consumption (SEC) rises linearly with load fraction l = P_el/C_el, reflecting lower overpotentials at part load [23]; compression to storage pressure adds a constant 2.5 kWh/kg:

$$\dot m_{H_2}(t) = \frac{P_{el}(t)}{SEC(l) + e_{c}}, \qquad SEC(l) = 48 + 6\,l \;\; [\mathrm{kWh/kg}], \qquad 0.1\,C_{el} \le P_{el}(t) \le C_{el} \;\text{or}\; P_{el}(t)=0$$

At rated load SEC = 54 kWh/kg, a system efficiency of about 62 % on the lower heating value (33.3 kWh/kg). Water use is 15 L per kg H_2 including purification losses.

3.3 Battery and hydrogen storage

$$E(t+1) = E(t) + \eta_b\,P_{ch}(t) - P_{dis}(t)/\eta_b, \quad 0.1\,C_b \le E(t) \le C_b, \quad P_{ch}, P_{dis} \le 0.5\,C_b$$

$$H(t+1) = H(t) + \dot m_{H_2}(t) - \min\bigl(H(t) + \dot m_{H_2}(t),\, D\bigr), \quad 0 \le H(t) \le C_{H}$$

The tank starts half full; any end-of-year deficit below the initial level counts as shortfall, so a design cannot "borrow" hydrogen from the next year.

3.4 Rule-based energy management

Each hour the controller (i) caps electrolyser power at the lesser of its rating and the power needed to fill the tank; (ii) serves it from PV first and tops up from the battery; (iii) idles the electrolyser if the available power is below minimum load; (iv) charges the battery with any PV left over and curtails the rest. This priority logic mirrors the battery-aware task scheduling of an autonomous agent: keep the high-value converter busy, and spend stored energy only where it unlocks production.

3.5 Economics

All capital costs are annualised with the capital recovery factor CRF(r, n) = r(1+r)^n / ((1+r)^n − 1) at r = 7 %. The levelised cost of hydrogen for weather scenario s counts every delivered kilogram, whether produced on site or bought in:

$$LCOH_s = \frac{\sum_{k} C_k\,c_k\,[CRF(r,n_k) + o_k] \; + \; c_w\,W_s \; + \; c_{bk}\,U_s}{365\,D}$$

where C_k are the four capacities, W_s the water volume, U_s the unmet hydrogen (kg) and c_bk = 10 USD/kg the backup price. Supply reliability is ρ_s = 1 − U_s / (365 D).

Component	Capital cost	Fixed O&M	Life (yr)	Search range
PV array	650 USD/kWp	15 USD/kWp·yr	25	100–4 000 kWp
PEM electrolyser (incl. BoP, compressor)	1 100 USD/kW	3 %/yr	20	50–2 000 kW
Li-ion battery	280 USD/kWh	2 %/yr	12	0–4 000 kWh
Compressed H_2 tank	500 USD/kg	1 %/yr	25	10–3 000 kg
Water	3.0 USD/m³ (arid), 1.5 USD/m³ (humid)	–	–	–

Component	Capital cost	Fixed O&M	Life (yr)	Search range
Backup hydrogen	10 USD/kg	–	–	–

Cost figures are indicative 2025-level assumptions chosen from the ranges reported in [1, 9]; Section 6.4 tests their influence.

4 Data and the Generative Scenario Model

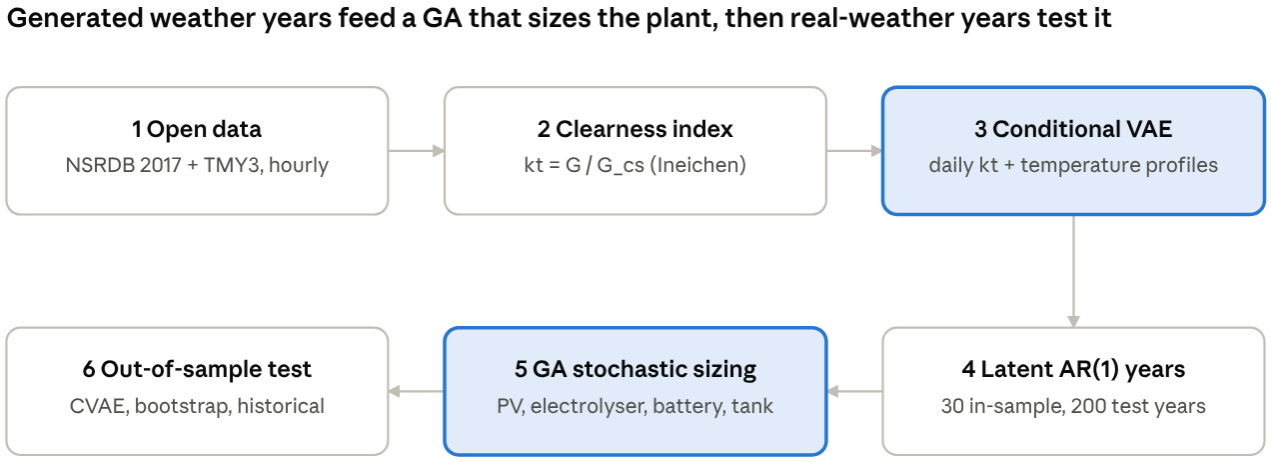


Fig. 1 · end-to-end workflow of the chapter · 6 stages

The generator (stage 3) is trained once for both sites; the GA (stage 5) never sees the weather it is finally judged on (stage 6).

4.1 Open datasets

Two open NREL datasets, mirrored in the pvlib-python repository [17], represent contrasting climates. Both were converted to hourly GHI and air temperature, and saved with the derived variables as a CSV/Excel workbook that accompanies the chapter.

Site	Source	Location	Elevation	Resolution	Annual GHI	Mean air temp.
A – semi-arid high desert	NSRDB PSM3, year 2017 [18]	40.53° N, 108.54° W (Colorado, USA)	2 168 m	30 min → 1 h	1 749 kWh/m²	8.1 °C
B – humid subtropical	TMY3 station 723170 [19]	36.10° N, 79.95° W (Greensboro, NC, USA)	273 m	1 h	1 566 kWh/m²	14.4 °C

Site A stands for the dry, high-irradiance, cold-winter locations where many hydrogen hubs are planned; site B for humid regions with frequent convective cloud. Every fifth day (73 per site, spread over all seasons) is held out for testing; the remaining 292 days per

site (584 in total) train the model.

4.2 Physics-informed representation

Rather than learning raw irradiance, the model learns the hourly clearness index $k_t = G / G_{cs}$, where G_{cs} is the Ineichen clear-sky irradiance [20] computed from the sun position and Linke turbidity. Generated irradiance is recovered as $G = k_t \cdot G_{cs}$ with k_t clipped to $[0, 1.3]$. Night hours are therefore exactly zero, the seasonal envelope and day length are exact, and the network only has to learn the weather — the part that is genuinely uncertain. Each training sample is a 48-value vector: 24 hourly k_t values and 24 standardised temperatures (scaled by 0.5 so irradiance dominates the loss).

4.3 Conditional variational autoencoder

The CVAE [4] has an encoder $q_\varphi(z \mid x, c)$ and a decoder $p_\theta(x \mid z, c)$. The condition $c = [\sin(2\pi d/365), \cos(2\pi d/365),$ site one-hot] tells both networks the day of year d and the site, so one model serves both climates. The encoder has one hidden layer of 96 tanh units and outputs the mean and log-variance of an 8-dimensional latent z; the decoder has two hidden layers of 96 tanh units. Training minimises the negative evidence lower bound with a Gaussian likelihood ($\sigma^2 = 0.01$) and a KL weight β annealed from 0 to 1 over the first 600 epochs:

$$\mathcal{L}(\theta,\phi) = \mathbb{E}_{q_\phi(z\mid x,c)}\Bigl[\tfrac{1}{2\sigma^2}\lVert x - \hat x_\theta(z,c)\rVert^2\Bigr] + \beta\, D_{KL}\bigl(q_\phi(z\mid x,c)\,\Vert\,\mathcal{N}(0,I)\bigr)$$

The network, back-propagation and Adam optimiser [21] are implemented from scratch in NumPy (no deep-learning framework), and the analytic gradients were verified against finite differences (maximum relative error 2.6×10^{-9}). Training uses 3 000 epochs, batch size 64 and learning rate 10^{-3}.

4.4 From days to years: latent AR(1) persistence

Sampling each day independently would destroy multi-day weather persistence — the cloudy spells that empty a storage tank. We therefore encode all 365 observed days, estimate the lag-1 autocorrelation ρ of the latent means, and generate a year by an AR(1) chain in latent space, which keeps each z_d marginally standard normal:

$$z_{d+1} = \rho\, z_d + \sqrt{1-\rho^2}\;\varepsilon_d, \qquad \varepsilon_d \sim \mathcal{N}(0, I)$$

Decoding z_d with the condition of day d gives one synthetic year of 8 760 hourly values. Thirty such years per site form the in-sample set for optimisation; 200 fresh years form the out-of-sample test set.

4.5 Baselines and scores

The CVAE is compared with (i) a month-wise multivariate Gaussian fitted to the same 48-value vectors, (ii) a seasonal bootstrap that resamples training days within ±15 days of the target date, and (iii) a deterministic climatology. For each held-out day, 100 samples are scored with the energy score (ES) on the 24-h GHI profile, the CRPS of daily insolation [16], and the coverage of the central 90 % interval (PICP90; ideal 0.90).

5 Stochastic Optimisation with a Genetic Algorithm

5.1 Formulations

The decision vector is x = (C_pv, C_el, C_b, C_H): PV capacity (kWp), electrolyser rating (kW), battery energy (kWh) and tank size (kg). It is a two-stage problem: x is fixed before the weather is known, and the hourly dispatch is the recourse decided by the energy management system of Section 3.4. Three formulations are compared:

1. **Deterministic (D)** — minimise LCOH on the single historical year, with reliability ⩾ 95 % in that year.
2. **Risk-neutral stochastic (RN)** — minimise the expected LCOH over S = 30 CVAE years.
3. **Risk-averse stochastic (RA)** — minimise an equal blend of expected LCOH and its CVaR at the 90 % level [12], so the worst 10 % of years also count.

Both stochastic formulations carry a chance constraint: at least 90 % of scenarios must reach 95 % on-site supply reliability. It is handled with a penalty, so the GA works on one scalar fitness:

$$\min_{x \in X} \; F(x) = (1-\lambda)\,\frac{1}{S}\sum_{s=1}^{S} LCOH_s(x) + \lambda\, \mathrm{CVaR}_{0.9}\bigl[LCOH(x)\bigr] + \mu \max\Bigl(0,\; 0.9 - \frac{1}{S}\sum_{s=1}^{S} \mathbb{1}[\rho_s(x) \ge 0.95]\Bigr)$$

with λ = 0 (RN) or 0.5 (RA), μ = 20 USD/kg, and box bounds X from the search ranges in Section 3.5.

5.2 Genetic algorithm

The plant simulator is non-linear and non-smooth (minimum load, part-load SEC, storage limits, rule-based logic), so a gradient-free GA [10] is used. It extends the operators of a discrete task-assignment GA to real-valued sizing:

- **Encoding:** a real vector of the four capacities, initialised uniformly within the bounds.
- **Selection:** tournament of size k = 3.
- **Crossover:** BLX-α (α = 0.5) with probability 0.9, which can create children slightly outside the parents' range and so keeps exploring.
- **Mutation:** Gaussian, per gene with probability 0.2 and standard deviation 10 % of the range, clipped to the bounds.
- **Elitism:** the two best designs pass unchanged.
- **Budget:** population 40, 60 generations (2 320 fitness evaluations).

Each fitness evaluation simulates every design in the population against every scenario at once: the simulator is vectorised over a (designs × scenarios) array, so one generation runs 40 × 30 × 8 760 ≈ 10.5 million plant-hours in under a second on one CPU core. As a sanity check the GA is compared with pure random search at the same budget over three seeds.

5.3 Out-of-sample evaluation

A design that looks good on its own scenarios may be over-fitted to them. Every design is therefore re-simulated on three sets it never saw: 200 fresh CVAE years, 200 seasonal-bootstrap years built from the observed data, and the historical year itself. The bootstrap set is the referee: it contains only real days, so a design that does well there is not exploiting artefacts of the generator.

6 Case Study Results

On real-weather test years, the GA designs sized on generated scenarios meet the 95 % reliability target far more often than designs sized on one historical year, for an LCOH premium of only 1–2.5 %. All results come from one run of the released code (CVAE training: 29.5 s; each GA run: 20–45 s on one CPU core).

6.1 Quality of the generated scenarios

The CVAE matches or beats every baseline on all three scores at both sites, and is the best calibrated: its 90 % intervals cover 88 % and 86 % of held-out days, against 75 % and 85 % for the bootstrap. The margin over the bootstrap is small on the energy score, which is expected — resampling real days is a strong baseline — but the CVAE can also produce days that never occurred and can be steered by its condition.

Site	Method	Energy score (kW/m²) ↓	CRPS daily insolation (kWh/m²) ↓	PICP90 (ideal 0.90)
A semi-arid	**CVAE (this work)**	**0.315**	**0.684**	**0.88**
A semi-arid	Seasonal bootstrap ±15 d	0.316	0.714	0.75
A semi-arid	Monthly Gaussian	0.316	0.719	0.81
A semi-arid	Climatology (point)	0.722	2.129	–
B humid	**CVAE (this work)**	**0.322**	**0.694**	**0.86**
B humid	Seasonal bootstrap ±15 d	0.324	0.694	0.85
B humid	Monthly Gaussian	0.330	0.716	0.85
B humid	Climatology (point)	0.610	1.618	–

Scored on 73 held-out days per site with 100 samples per day.

At the year level, the latent AR(1) coefficient is ρ = 0.35 (site A) and 0.30 (site B). Synthetic years reproduce the observed annual insolation (A: 1 761 vs 1 749 kWh/m²; B: 1 589 vs 1 566 kWh/m²) with a 5–95 % range of about ±3 %. The longest run of days below half the clear-sky energy averages 3.7 days for site A against 4.0 for bootstrap years, and 4.1 against 3.8 for site B; so at the arid site the generator slightly under-represents long overcast spells. Tilting the array at latitude raises the specific yield of the historical year from 1 435 to 1 587 kWh/kWp (A) and from 1 279 to 1 359 kWh/kWp (B).

6.2 Optimal designs

Site	Formulation	PV (kWp)	Electrolyser (kW)	Battery (kWh)	H_2 tank (kg)	CAPEX (M USD)	Mean LCOH, bootstrap years (USD/kg)	Mean reliability, bootstrap years	Years with reliability ≥ 95 %
A semi-arid	Deterministic	1 778	876	86	213	2.25	7.69	94.6 %	32 %

Site	Formulation	PV (kWp)	Electrolyser (kW)	Battery (kWh)	H_2 tank (kg)	CAPEX (M USD)	Mean LCOH, bootstrap years (USD/kg)	Mean reliability, bootstrap years	Years with reliability ≥ 95 %
A semi-arid	Risk-neutral	1 780	906	73	224	2.29	7.77	95.0 %	49 %
A semi-arid	Risk-averse	1 813	905	64	187	2.29	7.77	95.0 %	50 %
B humid	Deterministic	1 897	892	90	181	2.33	7.90	94.8 %	42 %
B humid	Risk-neutral	2 061	866	59	257	2.44	8.10	95.7 %	78 %
B humid	Risk-averse	2 121	836	80	241	2.44	8.11	95.6 %	76 %

All metrics evaluated on 200 seasonal-bootstrap years that the optimiser never saw.

Three patterns stand out. First, the deterministic design meets the target exactly in its own year (95.0 %) but fails it in most real-weather years: it is tuned to one weather sample. Second, the stochastic designs buy robustness mainly with more PV and a larger tank (site B: +164 kWp and +76 kg), lifting the share of compliant years from 42 % to 78 % for a 2.5 % LCOH increase. Third, the risk-averse design is almost identical to the risk-neutral one: the LCOH spread across years is narrow ($CVaR_{0.9}$ exceeds the mean by only 0.2 USD/kg) because shortfalls are covered at a backup price close to the LCOH, so the tail carries little extra cost.

At the arid site the stochastic design reaches the chance constraint on CVAE years (P = 0.90) but only 0.49 on bootstrap years. This gap traces back to the slight under-representation of long overcast spells noted in Section 6.1: a generator that is a little too optimistic in the tail produces a design that is a little too small. At the humid site, where the generator's tail matches the data, CVAE and bootstrap results agree closely (0.81 vs 0.78).

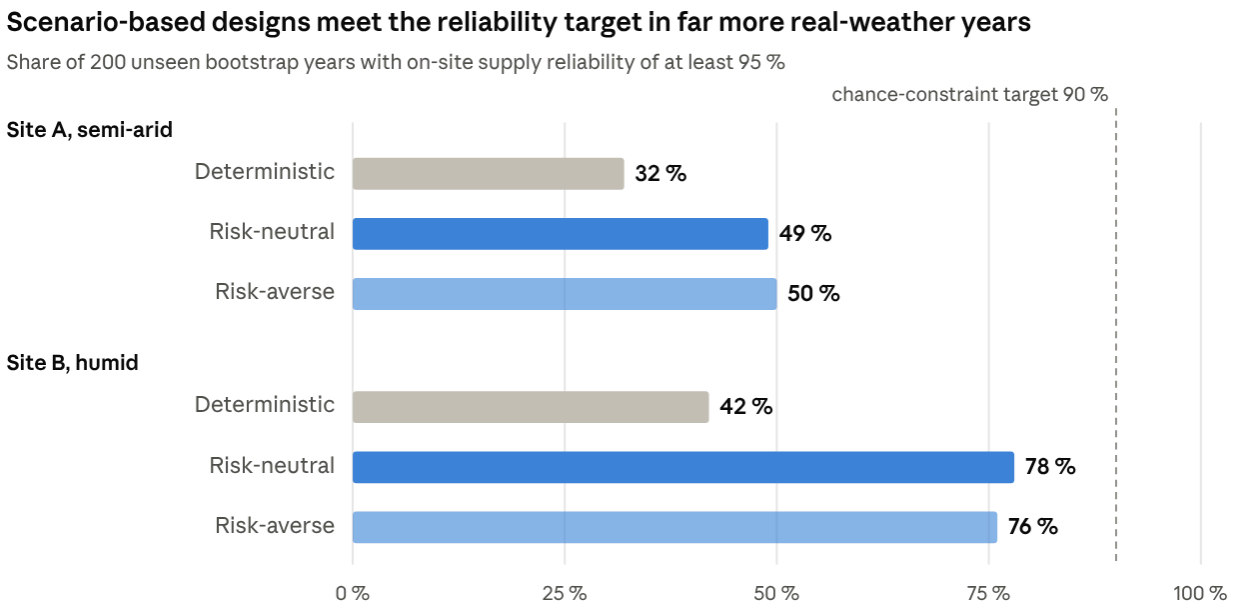


Fig. 2 · results/designs.csv (boot_p_rel95) · 200 bootstrap years per site

No design reaches the 90 % target on real-data years at site A, which points to the generator's tail rather than the optimiser.

6.3 Optimiser performance

With the same budget of 2 320 evaluations, the GA reached 7.63–7.64 USD/kg at site A and 8.07–8.08 USD/kg at site B across three seeds, a spread under 0.2 %. Random search ended at 7.92–8.80 and 8.50–8.96 USD/kg. The GA's elitism and BLX-α crossover concentrate the search near the cost-optimal ridge after about 600 evaluations.

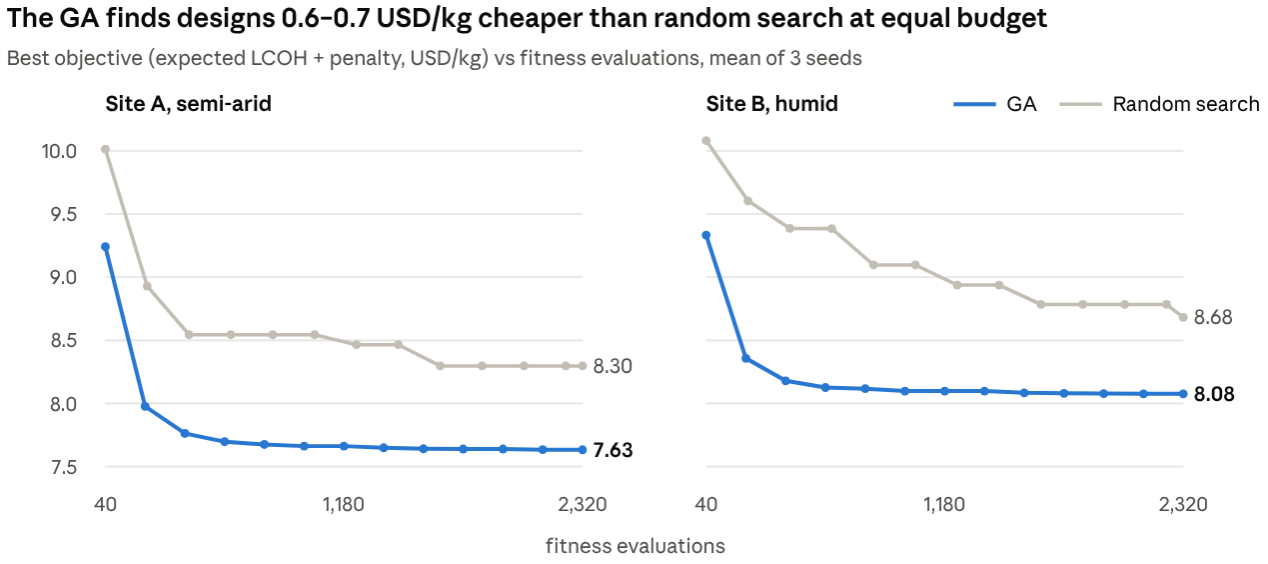


Fig. 3 · results/results.json (convergence) · risk-neutral problem, 3 seeds per method

6.4 Operation and sensitivity

The hourly trace of the risk-neutral design (figure file fig_dispatch.png in the code release) shows the electrolyser following PV to its rating at midday and then dropping to about 220 kW in the afternoon once the tank is full, matching the 4.2 kg/h demand. The tank cycles between about 78 % and 100 % every day and carries the whole night. The battery is almost unused: at 0.5 C, a 73 kWh battery can deliver only 37 kW, below the electrolyser's 91 kW minimum load, so it can never run the electrolyser at night on its own. Storing energy as compressed hydrogen (about 9 USD per kWh of electricity-equivalent at 500 USD/kg) is far cheaper than in batteries (280 USD/kWh), which is why the GA keeps the battery small. Electrolyser capacity factor is 25–27 % and 31–33 % of PV energy is curtailed in every design.

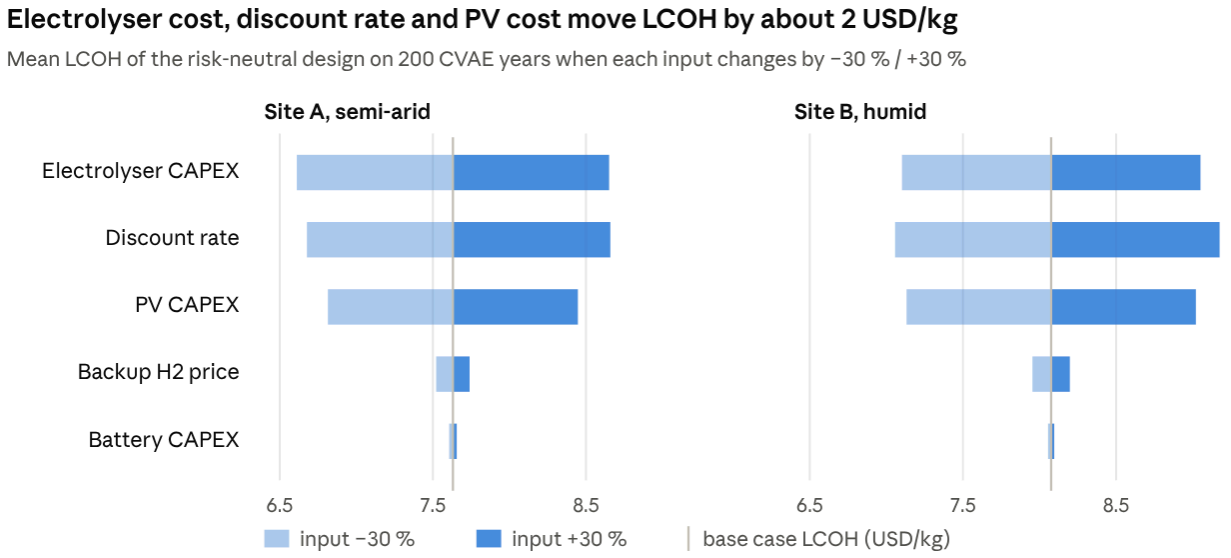


Fig. 4 · results/sensitivity.csv · base LCOH 7.63 (A) and 8.08 (B) USD/kg

The discount rate and the electrolyser and PV capital costs dominate LCOH; battery cost and backup price barely matter at these designs. Site B reacts slightly more strongly than site A to every parameter because it needs more installed capacity per kilogram.

7 Discussion: Operational Trends and Technical Challenges

The central lesson is that a design is only as robust as the weather it is optimised against — and that a generative model is useful exactly to the extent that its tails are right.

Design to a distribution, not a year. A single historical year produced a plant that met its reliability target in only 32–42 % of real-weather years. Scenario-based design raised this to 49–78 % for a 1–2.5 % cost premium. For project developers and lenders, who care about the bad years, this is a cheap insurance policy. The same workflow can serve as a stress test: generate a thousand years, rank them, and hand the worst to the engineers.

Validate the generator against real data, in the units that matter. The CVAE won on forecast scores, yet at the arid site it slightly under-produced long overcast spells, and the design inherited that optimism. Generic scores (energy score, CRPS) are necessary but not sufficient; a generator for storage sizing must also be checked on persistence statistics such as run lengths of low-irradiance days. A practical safeguard, used here, is to referee every design on bootstrap years built from real days.

Hydrogen storage, not batteries, buffers a constant off-take. The GA consistently chose small batteries and tanks of 190–260 kg. A battery's power rating must exceed the electrolyser's minimum load before it can run it overnight; below that, it only smooths start-ups. This interaction between minimum load and storage power is easy to miss in linear models that ignore minimum load, and it is captured naturally by a simulation-based GA.

Arid versus humid deployment. The semi-arid site gives 17 % more specific PV yield and a 0.3 USD/kg lower LCOH even with water priced twice as high, because water is a small cost item (15 L/kg). In real desert hubs, other harsh-environment factors — dust soiling, high cell temperatures, sand ingress into electrolyser balance-of-plant, and water sourcing from desalination — would need explicit modelling; the open-data workflow here is designed to be extended with such terms.

Trustworthy and responsible AI. A generative model in an engineering decision loop must be transparent and auditable. The design choices here support that: a physics-informed output space (clearness index on a clear-sky envelope), a small network written without a deep-learning framework, verified gradients, explicit baselines, out-of-sample tests on real data, and fully open data and code. Training took 30 s on one CPU core, so the computational footprint is negligible next to the infrastructure it informs.

Limitations. (i) Only one year of data per site was available offline; with multi-decade NSRDB or reanalysis data the generator could learn inter-annual variability directly. (ii) Day-to-day persistence uses a single AR(1) coefficient; regime-switching or sequence models (transformers, diffusion over weeks) would capture long spells better. (iii) The electrolyser model is static: degradation, stack replacement, ramp limits and cold-start losses are omitted. (iv) Cost data are indicative, and the demand is constant. (v) The GA uses a penalty for the chance constraint; multi-objective GAs such as NSGA-II [11] would expose the full cost–reliability trade-off.

Future directions. Promising next steps include conditioning the generator on climate-model projections for climate-aware design; replacing the hour-by-hour simulator with a neural-operator surrogate [24] to speed up large GA searches; degradation-aware model-predictive control for operation; and large language model agents that query the generator and optimiser to answer planners' "what if" questions in plain language.

8 Conclusion

This chapter linked a physics-informed conditional VAE to genetic-algorithm sizing of an off-grid solar hydrogen plant, using open NREL data for a semi-arid and a humid site. The CVAE, written from scratch in NumPy and trained in 30 s, matched or beat Gaussian, bootstrap and climatology baselines and gave the best-calibrated intervals. Designs optimised on generated weather years kept 95 % supply reliability in 49–78 % of unseen real-weather years, against 32–42 % for designs sized on one historical year, at an LCOH cost of 1–2.5 % (7.8–8.1 USD/kg at 100 kg/day). The GA outperformed random search by 0.3–0.9 USD/kg at equal budget, and showed that compressed-hydrogen storage, not batteries, should buffer a constant off-take. The main caution is that a generator's tail behaviour flows straight into the design, so generators for storage sizing should be validated on persistence statistics against real data. Code and data accompany the chapter so that every number can be reproduced.

Code and data availability. All code, the processed datasets (CSV and Excel) and the result files are in the Book-Chapter-Green-Hydrogen folder of the authors' repository.

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